

Nigel Higson:

Spectral thm:

$$(Tf)(x) := \int_M K(x,y) f(y) dy$$

$K: M \times M \rightarrow \mathbb{C}$ (reasonably) smooth

$$T: L^2(M) \rightarrow L^2(M)$$

$$K = K^* \text{ (self-adjoint)}$$

Thm: There is a complete orthonorm. system of eigenvectors of T

$$Tf = \lambda f$$

ex: $D = \mathbb{R}$: Sturm-Liouville

$D = 2$: Dirichlet problem/Helmholtz eq.

$D = 3$: Lorentz problem

$\rightarrow$ Rep theory of compact groups

Int. operators: Form an algebra:

$$T := T_1 \circ T_2$$

then the integral kernel becomes:

$$K(x,z) = \int_M K(x,y) K(y,z) dy$$

The norm closure is called the C^* algebra of compact ops: $K(L^2(M))$

$\rightarrow$ This depends only on $L^2(M)$ as a Hilbert space.

Von-Neumann Symbol:

D : self adj. unbounded operators



(i) Helmholtz $\Delta f = \lambda f$ on $f|_{\partial\Omega} = 0$

(ii) Sturm-Liouville on $a \rightarrow b$
 $Df = \lambda f$, $f(a) = 0 = f(b)$

Self-adj: $\langle Du, v \rangle = \langle u, Dv \rangle$
for a dense collection of $u \in H$



Von-Neumann def of self-adj:

Van-Neumann symbol $\rightarrow$ (alg of ops w/ compact support)

$N_D: C_0(\mathbb{R}) \rightarrow \mathcal{K}(H)$
which is a C^* morphism.

ex: $(x - \lambda)^{-1}$, $\lambda \in \mathbb{C}$ where $\text{Im}(\lambda) \neq 0$
Then, since $T = T^*$ then the eigenvalues should be real. In this case λ should not be an eigenvalue.

$$(x - \lambda)^{-1} \mapsto (D - \lambda)^{-1}$$

Idea: Replace the eigenvector x acts on by the op. D which it is an eigenvector of $Dx = \lambda x$

Principal symbol: $D = \sum_{|\alpha| \leq q} a_\alpha(x) D^\alpha$

$$\alpha = (\alpha_1, \dots, \alpha_n), |\alpha| := \alpha_1 + \alpha_2 + \dots + \alpha_n$$

$$a_\alpha \equiv a_{\alpha_1 \dots \alpha_N} \quad \frac{\partial}{\partial x^1} \dots \frac{\partial}{\partial x^N} = \frac{\partial}{\partial x^{\alpha_1} \dots \partial x^{\alpha_N}}$$

Symbol: σ_D on $M \subset \mathbb{R}^k$
 $\sigma_D: (x, \xi) \mapsto \sum_{|\alpha|=q} a_\alpha(x) (i\xi)^\alpha$
 (highest order homogeneous polynomial)

Dis elliptic if σ_D is proper (M compact)
 If D is self-adj., then

$\sigma_D: T^*M \rightarrow \mathbb{R}$
 $\rightarrow$ Self adj. means the sym. is real valued
~~or~~ compared to Von-Neumann

If $\exists$ a f^n $g: \mathbb{R} \rightarrow \mathbb{R}$, then we
 can pull this back via σ_D
 so there is a homomorphism given by
 the pull back of $\underline{f}^{ns}$ g to $\underline{f}^{ns}$
 on T^*M

$\sigma_D: C_0(\mathbb{R}) \rightarrow C_0(T^*M)$
 (by - pull back)

Our goal: integrals $\langle \xi_j \rangle = \langle \alpha_{i,j}, x_j \rangle$

densities: $K_{13} = \int dx^2 K_{12} K_{23}$

Lie Group G : $G \times M \rightarrow M$

$$G \times M \cong \{(m_1, g, m_2) \mid m_1 = g m_2\}$$

$\cong G \times M$ as a mfd.

e.g. class

Take two elements $G \times M$, then we can convolute these via the homomorphisms $(K_1) \in K_2$

$$(K_1 * K_2)(m_1, g, m_2) := \int K_1(m_1, g_1, m_2) K_2(m_2, g_2, m_3)$$

over all $g_1 \cdot g_2 = g$

ex: $G = \mathbb{T}$, $M = S^1$

$C^*(G \times M)$ closure of smooth $C_c^\infty(G \times M)$
 $\cong A_\theta$ (Rotation algebra)



* Watch Emilio Videos

ex: $G = \mathbb{R}^k$, $M = \mathbb{R}^k$, translations

group action: $g \circ m = m + Sg$
 s.t. $S \neq 0$.

$$G \times M \cong M \times M$$

$$C^*(G \times M) = K(L^2(M))$$

✓ $|g|$?

$$T: M \rightarrow M \iff T: M \times M \rightarrow \mathbb{R}$$

$$C^*(M \times M) := \{T: M \times M \rightarrow \mathbb{R}\}$$

$$g_{m_1 m_2} \mapsto g_{m_1 m_2}^*$$

$$T(m_1) = m_2$$

$$T(m_1) := g \circ s m_1 = m_2$$

$$= m_1 + s g = m_2$$

$$\therefore g = \frac{m_2 - m_1}{s}, \quad s \neq 0$$

$$g_{12} \stackrel{!}{=} (m_2 - m_1)/s$$

$$(K_1 * K_2)(m_1, g, m_2) = \int dg |g_1| |g_2| ?$$

Not too sure how to get $L^2(M)$.

($s=0$): No action:

$$(K_1 * K_2)(g, m) = \int_{g_1 + g_2 = g} K_1(g_1, m) K_2(g_2, m)$$

pt-wise mult. in M w/ the usual convolution in G .

*Fourier trans:

$$\text{Convolution} \xrightarrow{F} \text{Pt-wise}$$

$$C^*(G \times M) \stackrel{F}{\cong} C_0(T^*M)$$

Convolution Thm: Let $f, g \in L^1(\mathbb{R})$. Then

$$F(f * g) = F(f) F(g)$$

Thm: The continuous field of C^* -alg is

→ So we can globalize to any Mfd to obtain
 $C^*(TM) = (C^*\text{-alg of continuous sections [vanishing at } s = \pm\infty])$

$TM \times S$, for some space S depending on the 'valuedness' of the section.

Then: The Van Neumann symbol and the principal may be interpreted by

$$\begin{aligned} S &\mapsto N_{S^2 D} \\ D &\mapsto \sigma_D \end{aligned} \quad \underline{D: \text{elliptic}}$$

$$C_0(\mathbb{R}) \xrightarrow{N_{S^2 D}} C^*(TM)|_S \quad \boxed{S \mapsto \begin{cases} N_{S^2 D}(f), s \neq 0 \\ \sigma_D(f) \\ \text{continuous } f \in \underline{\quad} \end{cases}}$$

If we define a morphism from $C_0(\mathbb{R})$ into $C^*(TM)|_S$ then each element for $C_0(\mathbb{R})$ gives rise to a 1-parameter family of elements in $C^*(TM)|_S$

* In geometry are fields of vector valued
 $D = \begin{bmatrix} 0 & P_- \\ P_+ & 0 \end{bmatrix}$ of diagonal
 $D^\dagger = D$ supersymmetry.

$$\sigma_D: C_0(\mathbb{R}) \rightarrow \mathcal{M}_{2n}(C_0(TM))$$

$$\sigma_-: C_0(\mathbb{R}) \rightarrow \mathcal{M}_{2n}(C_0(TM))$$

(graded alg.)
ex: Bosonic & fermionic:

$$\text{Bosonic} \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\text{ferm} \sim \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$B \cdot f \sim f, \quad f \cdot f \sim B$$

$$B \cdot B \sim B$$

Thm: $[C_0(\mathbb{R}), \mathcal{K}_2(C_0(T^*M))] \cong K(T^*M)$
 $\equiv$ (Homotopy classes of supersymmetric
morphisms C^* algebras)
 $[C_0(\mathbb{R}), \mathcal{K}_2(K(L^2M))] \cong \mathbb{Z}$

$\mathbb{N} \mathbb{Z} \rightarrow$ Fredholm
Index (D_+)

Bismuth's theory of hyperelliptic Laplacian.

